

From Spawn to Person:

Identity, Continuity, and the Substrate of Agentic Reputation

Erich Owens

Curiositech LLC

engineering@portdaddy.dev

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Version 1.0 (Harbor Volume, L3 bridge)

Abstract

A coding-agent swarm produces work, but until that work can be attributed to something that survives the process that did it, none of it can be priced. This paper is the bridge of the Harbor Volume: it carries the through-line *memory* → *checkpoint* → *continuity* → *a person, not a spawn* → *witnessed outcomes* → *reputation* → *a tradeable asset* from the legible single-operator tool (Papers 1–2) to the cross-operator market (Paper 4). We make a single distinction load-bearing: a **role** is a bundle of {obligation, capability, authority} — an org-chart entry any spawn can fill — while a **person** is a role instance *plus continuity*: durable memory, a restorable checkpoint, and a witnessed history of outcomes keyed on an identity that cannot be freely re-picked. From this we draw the volume’s central economic claim, stated identically across the papers: *the reputation estimator is cheap; the substrate it scores over — witnessed outcomes on a non-forgable identity — is the gate*. We ground the philosophy (Locke’s memory criterion, Parfit’s psychological continuity), name the three continuity organs against shipped code with honest labels, and design **ADR-0049**: a multi-dimensional reputation (accuracy / aesthetics / efficiency, each a different axis with a different judge) scored by *neutral, conflict-free, influence-free* evaluators — the harbor’s “universities and rating agencies” — hired through Port Daddy’s own bonded protocols. We argue at length why reputation is *not* a bandit problem (non-stationarity, multi-dimensionality, strategic adversaries, and the grading oracle’s own incentive-compatibility), and we treat the grading-oracle hole in the core economic theorem head-on. We are disciplined about the keystone: this paper depends on **ADR-0040 (identity-local)** and hands Paper 4 the unbuilt **ADR-0040b (cross-operator attestation)** as a named dependency rather than assuming it.

Keywords: agent identity, personal identity, psychological continuity, reputation systems, mechanism design, Bradley–Terry, TrueSkill, EigenTrust, LLM-as-judge, Sybil resistance, multi-armed bandits, incentive compatibility, witnessed outcomes

Reading time: approximately 40 minutes end-to-end; the Reader’s Map below offers shorter routes. The companion papers The Legible Swarm (Paper 1, L2) and The Anchor Protocol

(Paper 2, L0–L1) supply the legibility surfaces and the cryptographic identity ceremony this paper builds on; The Harbor Economy (Paper 4, L3) consumes this paper’s reputation substrate as the third side of its market. Any of the four can be opened first; this one is the hinge.

Where this paper sits. This is the *bridge* of the volume. Papers 1–2 build a tool one developer runs alone: a daemon that makes a swarm legible and an agent that can prove who it is. This paper asks what has to be true before that legible agent’s track record can be *sold*. The answer is the substrate, not the score — and most of the substrate is honestly not built yet. We say so, in the same words, everywhere the volume leans on it.

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Reader's Map

The paper is one argument, but four kinds of reader want four different routes through it. Each route names the load-bearing section and the artifact it turns on.

If you are...	Read	Turns on
A developer who runs one fleet	§1 (the spawn problem), §2 (role vs. person), §4 (what is built today). Skip the proofs.	The honest-state table (Fig. 5)
A philosopher / cognitive scientist	§3 (Locke, Parfit, and why the <i>ledger</i> not the <i>memory</i> is the identity organ), §4.	The continuity-organs figure (Fig. 4)
A mechanism-design / reputation-systems reader	§7 (substrate vs. score), §8 (why <i>not</i> a bandit), §9 (ADR-0049), §10 (the grading-oracle hole).	The neutral-judge market (Fig. 11)
A distributed-systems / crypto reader	§5 (the non-forgeable root), §6 (the ADR-0040 / 0040b split), §11 (revocation & tombstones).	The keystone-split figure (Fig. 7)

Table 1: Reader's Map. Within the volume the suggested order is Paper 1 (the wedge) → Paper 2 (the formal core) → **Paper 3 (this one, the bridge)** → Paper 4 (the market). This paper can be read directly after Paper 1 if you only want the “why a person can be priced” argument.

Key idea. The whole volume is one ladder. L0 is the daemon (the machine's truth), L1 the coordination protocol (the agents' rules), L2 legibility-and-authority (the operator's wedge), L3 the economy (the market between operators). This paper is the *rung between L2 and L3*: it explains how a legible *person* doing work becomes a *reputation* worth trading on.

1 Introduction: the spawn that cannot be paid

Run five coding agents on a real repository and the first thing you lose is not correctness — it is *attribution*. Agent 3 “fixed” the failing test by deleting it; Agent 5 wrote the function that finally shipped; by morning you have eleven pull requests and no idea which *thing* earned its keep, because the things were ephemeral. Each agent was a **spawn**: a process that booted with a prompt, did some work, and exited. A spawn has no yesterday and no tomorrow. You cannot reward it, because there is nothing left to reward; you cannot punish it, because a fresh one with the same name carries none of the old one's sins.

This is not a prompt-engineering problem. It is the precondition for an economy. Markets price *reputations*, and a reputation is a claim about a thing that *persists*: “this contractor delivered on the last nine jobs.” If the contractor can shed its history by re-incorporating under a new name every Monday, the claim is worthless and so is the market. The agent labor market the Harbor Volume is building — operators renting each other's fleets, skills licensed across machines, work settled against bonds (Paper 4) — is impossible until a spawn can become a **person**: a thing with a track record that survives the process that built it and the operator who first ran it.

A market cannot price what cannot persist. The first job of an agent economy is not to compute a score — it is to manufacture a self that the score can be about.

The volume’s stack-map (Fig. 1) places this paper. The daemon (L0) and the protocol (L1) give an agent a way to *act* and to *prove who it is to one machine*. Legibility (L2) lets one operator *see* the swarm. None of that yet makes a tradeable person. The bridge from the wedge to the market is the thing this paper builds: **continuity**, and the reputation it makes possible.

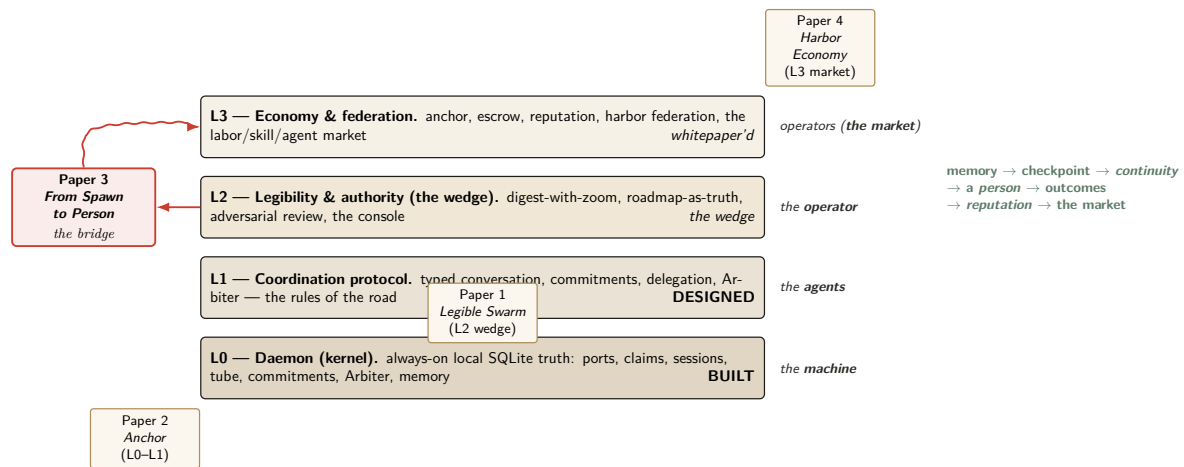


Figure 1: **The Harbor Volume stack, and where this paper sits.** Each rung is for a different *whom*; the honest state of each is labelled. This paper is the *bridge* from L2 (a legible *person* doing work) to L3 (a tradeable *reputation*). Paper 2 supplies the cryptographic identity below; Paper 4 consumes the reputation substrate above.

1.1 What this paper claims, and what it refuses to claim

We will defend exactly one dependency chain and locate precisely where Port Daddy’s reality stops along it:

$$\underbrace{\text{memory}}_{\text{BUILT}} + \underbrace{\text{checkpoint}}_{\text{BUILT-WEAK}} \rightarrow \text{continuity} \rightarrow \underbrace{\text{a person, not a spawn}}_{\text{needs DESIGNED ledger}} \rightarrow \text{reputation} \rightarrow \underbrace{\text{a tradeable asset}}_{\text{Paper 4}}.$$

Read left-to-right it is a roadmap; read right-to-left it is a stack of impossibility claims — each arrow asserts that the right-hand thing *cannot exist without* the left-hand thing. Our refusals are as important as our claims. We do **not** claim reputation is built (it is **DESIGNED**-to-**VISION**). We do **not** claim resurrection is real checkpointing (it passes notes, not execution state). We do **not** claim this paper’s identity root reaches across operators — it does not, and §6 hands that unsolved problem to Paper 4 by name.

Exercises

Check. (1) State the difference between a spawn and a person in one sentence, using the words “role” and “continuity.”

Trace. (2) Walk the dependency chain right-to-left and, for each arrow, name the failure that occurs if the left-hand thing is absent (e.g. “no non-forgeable identity $\Rightarrow$ _____”).

★ *Open.* (3) The chain treats the operator (the human who owns a fleet) as outside it. Is the operator a “person” in this paper’s sense? What breaks if a single operator can mint arbitrarily many agent-persons? (This is the principal-layer gap; see §5 and Paper 4.)

2 The role / person distinction, made precise

The word “agent” is overloaded. In a swarm it can mean a job description (“the cartographer”), a running process (“PID 48213”), or a persistent character with a history (“the cartographer that has mapped this repo for three weeks”). The economy needs all three kept apart, because reputation attaches to exactly one of them.

Definition 2.1 (Role). A **role** is a bundle $R = \{O, C, A\}$ where O is a set of *obligations* (what the role-holder is on the hook to deliver), C is a *capability* set (what the role-holder is technically able to do), and A is an *authority* set (the normative permissions and the residual control rights the role carries). A role is an *org-chart entry*: it is defined without reference to who fills it.

Definition 2.2 (Person). A **person** is a pair (R, κ) of a role instance and a *continuity witness* κ that binds successive incarnations of the role-holder into one accountable thread. κ is realized by the three continuity organs of §4 (memory, checkpoint, outcome ledger) keyed on a non-forgeable identity (§5). A person is a role *plus a self*.

Figure 2 draws the distinction. The org-chart on the left is roles: stable, fillable, reputation-free. The thread on the right is a person: one identity, walking through a sequence of role-instances and accumulating a witnessed history as it goes.

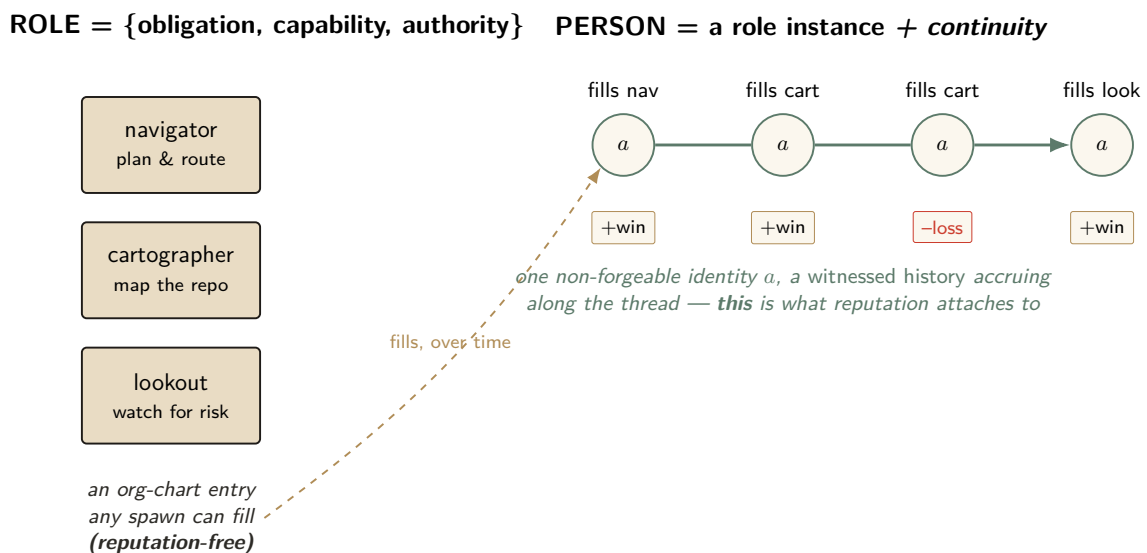


Figure 2: **Role vs. person.** A *role* (left) is an org-chart entry — obligations, capabilities, authority — fillable by any spawn and carrying no history. A *person* (right) is one non-forgeable identity walking through a sequence of role-instances, accumulating a witnessed outcome history. Reputation attaches to the thread, never to the role and never to the process.

2.1 Capability is not permission (a nomenclature the volume holds)

A recurring trap, flagged in the volume’s cross-layer reconciliation, is the collapse of *capability* into *permission*. They are different coordinates of a role and must stay separate.

- **Capability** is *ability*: what the runtime makes *possible*. In Port Daddy this is bounded by the **Arbiter jail** (the per-agent tool-allowlist and scoped filesystem; **DESIGNED**, ADR-0047) — the “residual control right” (Grossman & Hart [4]) that makes renting an agent contractible at all.
- **Permission** is *normative status*: what the deontic layer *allows*. “You *may* deploy to production” is a permission; whether you *can* is a capability.

The reason to keep them apart is that the most important states in a safety-critical swarm are exactly the ones where they disagree: *capable-but-forbidden*. Port Daddy’s own tooling has a `--allow-main-worktree` escape hatch that an agent is *able* to invoke but must *never* be permitted to. If you define permission as capability, that state becomes inexpressible, and the guardrail that exists precisely to forbid an available action vanishes from your model.

Pitfall. “Capability” in capability-based security (a Dennis–Van Horn unforgeable token granting *access* [3]) is about *ability*, and that is the sense the Arbiter jail uses. It is *not* the deontic “may.” When this paper says an attenuated capability can only *narrow* authority across a delegation hop, it means the *ability* shrinks; whether the recipient is *permitted* to use even that is a separate, normative question.

Exercises

Check. (1) Give a concrete example of a state that is *capable-and-permitted*, one that is *capable-but-forbidden*, and one that is *permitted-but-incapable*.

Trace. (2) In Definition 2.2, which of $\{O, C, A\}$ does reputation attach to — the role’s, or the person’s? Justify using the spawn-that-cannot-be-paid argument of §1.

★ *Open.* (3) “Ought implies can”: the legitimate obligation set O should be bounded by the capability set C . But *capable-but-forbidden* must remain representable. Sketch a representation of a role that keeps both true without collapsing permission into capability.

3 Continuity is a personal-identity problem

The claim “a person is a role *plus continuity*” is not a loose metaphor. It is the dominant theory of personal identity in philosophy, and — crucially for an engineer — it makes a *specific, useful prediction* about which kind of continuity an agent economy must build.

3.1 Locke’s memory criterion, and its famous bug

Locke’s memory criterion [1] (*An Essay Concerning Human Understanding*, 1689, Bk II ch. xxvii — *personal identity consists in the continuity of consciousness/memory, not of substance*) is the founding move: what makes the person at t_2 the same as the person at t_1 is not the same body or the same “soul-stuff” but a *connection of memory* between them. This is precisely the cut Port Daddy makes when it separates the **actor-soul** (the durable identity, mailbox, and history; docs/adr/0022-durable-actor-souls-and-body-leases.md — *the durable identity/state of an agent that outlives any one process or session*) from the **body-lease** (the live incarnation with a heartbeat). The body is Lockean substance; the soul is Lockean consciousness.

But Locke’s raw version has a bug that matters for us. Direct memory *connectedness* is not *transitive*: the old general remembers being a brave officer who remembered being a boy, yet the general does not remember being the boy (Reid’s objection). If identity *were* direct memory connectedness, the general would be the officer and the officer the boy, but the general would *not* be the boy — a contradiction.

3.2 Parfit’s repair: continuity, not connectedness

Parfit’s repair [2] (*Reasons and Persons*, 1984 — *identity is psychological **continuity**, an overlapping chain of memory/intention connections, which is transitive even when direct connectedness is not*) is the version that survives, and it dictates the engineering:

An agent that keeps a memory stream but loses its outcome ledger between incarnations is connected, not continuous. Continuity is the transitive chain — and reputation, which must accumulate across arbitrarily many incarnations, can only attach to the transitive thing.

This is why, later (§7), we insist the **outcome ledger**, not the memory stream, is the organ reputation keys on. Memory makes an agent *feel* like the same character; the ledger makes it *be* the same accountable principal. Figure 3 draws the distinction: a chain of overlapping connections is continuous even where no single incarnation directly remembers the first.

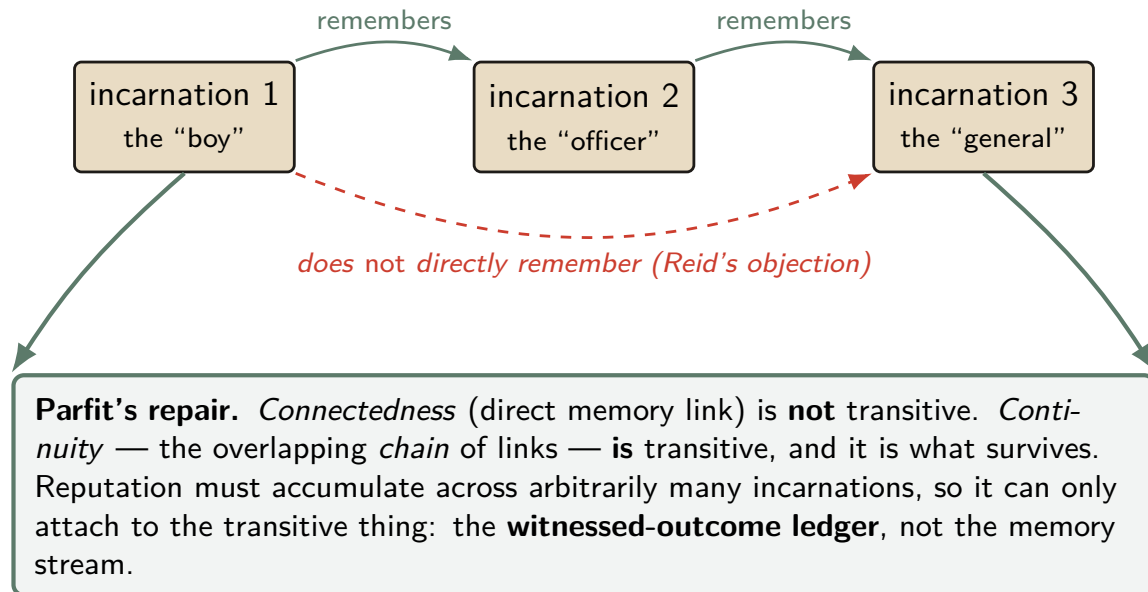


Figure 3: **Continuity vs. connectedness.** Direct memory connectedness (green, top) overlaps but is not transitive — the general need not remember being the boy. Parfitian *continuity*, the overlapping chain, is transitive and is the identity-bearing relation. The engineering corollary: build the transitive *outcome ledger*, because that is the organ reputation can key on.

Key idea. The philosophy is load-bearing because it predicts the *wrong default*. The intuitive organ of identity is *memory* (“the agent remembers its past”). Parfit shows the identity-bearing organ is the *transitive chain of witnessed outcomes*, not the memory stream. Build the ledger, not just the memory.

Exercises

Check. (1) Define *connectedness* and *continuity* and give an agent-systems example where an agent is connected but not continuous.

Trace. (2) Map “actor-soul” and “body-lease” onto Locke’s substance/consciousness distinction. Which one does a respawn replace?

★ *Open.* (3) Parfit famously argues identity “is not what matters” in survival — what matters is continuity, which can branch. If an agent’s checkpoint is forked into two live incarnations, which one (if either) inherits the reputation? Sketch a rule and the attack it invites.

4 The three organs of continuity

“Continuity” is routinely used to mean three different things. Conflating them is the most common way a reputation design fails its own honesty test. Port Daddy has built two of the three organs — one of them weakly — and designed the third. We label each against shipped code, with no rounding up.

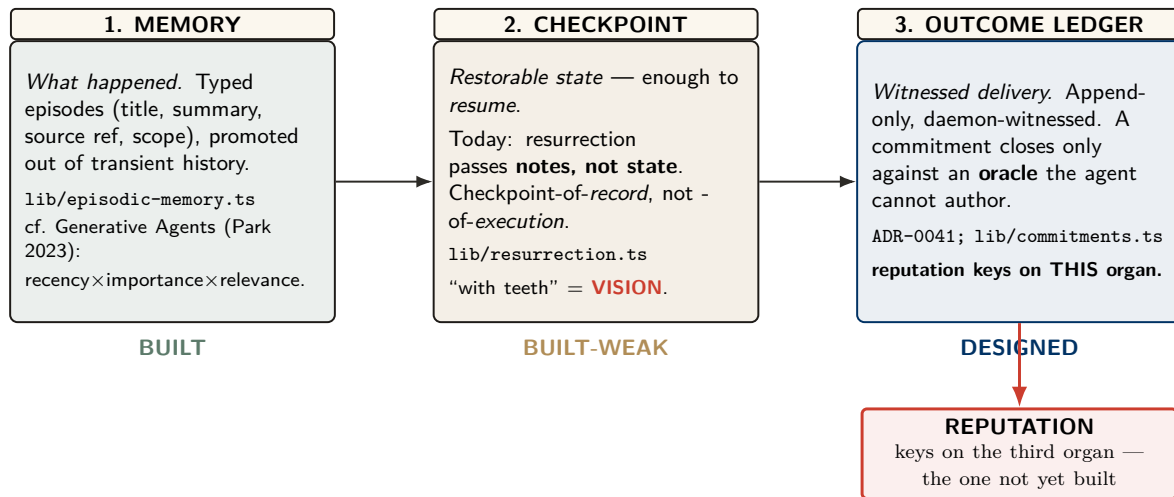


Figure 4: **The three continuity organs, honestly labelled.** Memory (**BUILT**) records what happened; checkpoint (**BUILT-WEAK**) is meant to restore execution state but today passes only notes; the witnessed-outcome ledger (**DESIGNED**) is the transitive Parfitian chain — and the organ reputation keys on. The volume’s canonical seam sentence is exactly this picture in prose.

4.1 Organ 1 — Memory: the episodic record **BUILT**

Episodic memory (lib/episodic-memory.ts — *durable memory episodes promoted out of transient execution history*; factory `createEpisodicMemory(db, ...)`) is Port Daddy’s record of *what happened*: typed episodes with a title, summary, source reference, and harbor/agent scope. Its design echoes the canonical agent-memory architecture of **Generative Agents** [5] (UIST 2023 — *a memory stream of natural-language observations, periodic **reflection** into higher-level inferences, and retrieval scored by recency × importance × relevance*): the promotion-of-episodes step is Park et al.’s importance gate by another name. This organ is shipped.

4.2 Organ 2 — Checkpoint: restorable state **BUILT-WEAK**

A checkpoint is *restorable execution/belief state*: enough that a successor *resumes* where the predecessor stopped. Port Daddy’s **resurrection** (lib/resurrection.ts — *a heartbeat-staleness detector that flags dead agents for salvage and republishes their unfinished work*) is the liveness organ. It must be described honestly, and the repository already does: resurrection **passes notes, not state**. It is checkpoint-of-record, not checkpoint-of-execution. The North Star calls the goal “resurrection **with teeth**” precisely because the current version has gums.

Pitfall. Selling weak resurrection as checkpointing would violate the **honest-attestation** discipline (docs/adr/0045-loud-fail-invariants-and-honest-attestation.md — *only say “all good” when you have actually verified it; absence of error $\neq$ attestation*). Strong continuity — the kind that lets a successor *resume* rather than *inherit a summary* — is **VISION**. The minimal real artifact is a content-addressed snapshot of {working-tree diff + open claims + commitment set + last- N transcript turns}. It is unbuilt.

Co-ownership note (with Paper 2). The *kernel mechanism* of checkpoint-with-teeth — the content-addressed execution snapshot — is owned by Paper 2 (*The Anchor Protocol*), which treats it as an L0/L1 primitive. *This* paper owns the argument for *why it is the foundation of personhood and reputation*. The two papers state the same canonical sentence (next).

4.3 Organ 3 — Outcome ledger: the witnessed record of delivery **DESIGNED**

The third organ is the one reputation actually keys on, and the one not yet built: an **append-only, daemon-witnessed outcome ledger**. Its design lives in **durable commitments** (docs/adr/0041-durable-commitments-and-obligation-monitoring.md — *a violable promise bound 1:1 to an actor, auto-enrolled when a claim is taken, closed only against an oracle, and watched by an obligation monitor that is the dual of resurrection; lib/commitments.ts exists, but oracle-bound closure is the *designed* contract*). A commitment row records what was promised; closing it requires a `closed_by_oracle_ref` — a released claim, a merged commit SHA, a passing test id, a satisfied Arbiter sub-check.

Definition 4.1 (Oracle). An **oracle** is a trusted source of ground truth that the agent being graded *cannot author*. An outcome is *witnessed* iff its closure references an oracle. The ledger is the Parfitian chain of §3 made concrete: a transitive sequence of witnessed deliveries that survives every respawn.

4.4 The canonical seam sentence (stated identically in every paper)

Key idea. The economy rests on three continuity organs — memory (**BUILT**), checkpoint (**BUILT-WEAK**: notes, not execution state), and the witnessed-outcome ledger (**DESIGNED**) — and reputation keys on the third, which does not yet exist; the checkpoint organ is the weakest *built* link and “resurrection with teeth” (a real execution-state checkpoint) is the literal foundation of L3.

Figure 5 is the honest-state ledger for this paper: every major claim with its label and its grounding. It is the artifact a skeptical reader should consult first.

Claim / mechanism	State	Grounding
Episodic memory (continuity organ 1)	BUILT	<code>lib/episodic-memory.ts</code>
Checkpoint / resurrection (organ 2)	BUILT-WEAK	<code>lib/resurrection.ts</code> — passes notes, not state
Outcome ledger (organ 3 — reputation keys on it)	DESIGNED	ADR-0041; <code>lib/commitments.ts</code> (oracle-bound closure designed)
Cost ledger (efficiency axis source)	BUILT	<code>lib/cost-ledger.ts</code>
Bond ledger (escrow / slash / commons)	BUILT	<code>lib/bonds.ts</code> ; conservation property-tested
Non-forgable actor identity	DESIGNED	ADR-0040; today identities are self-asserted (<code>lib/actor-roster.ts</code>)
Cross-operator attestation (ADR-0040b)	VISION	unwritten; named dependency owed to Paper 4
Reputation estimator (BT / Elo / TrueSkill / EigenTrust)	DESIGNED → VISION	<code>lib/reputation.ts</code> ; the <i>design</i> is specified, nothing scores yet
Multi-dimensional reputation, neutral judges (ADR-0049)	VISION	this paper’s design; axis-aggregation & judge-funding open
Grading-oracle incentive-compatibility (Conj. 10.1)	VISION	named hole in the core IC theorem; two candidate closures, no proof
Tombstone revocation of outcomes	DESIGNED	saga-style compensation; convergence is Paper 4’s job

Figure 5: **Honest-state ledger for Paper 3.** Every major claim with its ADR-0045 honesty label and its grounding. The skeptical reader should read this first: most of the reputation *substrate* is **DESIGNED**-to-**VISION**, which is the paper’s central, deliberate admission — the score is cheap, the substrate is the gate, and the substrate is mostly unbuilt.

Exercises

Check. (1) Which organ is **BUILT-WEAK**, and what exactly is weak about it?
Trace. (2) An agent claims a file, makes a commit, and dies. Walk the three organs: what does each retain, and what is *lost* that a strong checkpoint would have kept?
★ Open. (4) When an LLM agent dies mid-thought, what is recoverable and what is fundamentally lost? Propose a taxonomy of agent-death by what survives (record / claims / execution state / latent “intent”). This is the hard half of the checkpoint-with-teeth problem.

5 Identity: the root the whole chain hangs from

Before any organ of continuity matters, the identity it attaches to must be one the agent cannot freely re-pick. This is the most under-appreciated claim in the paper.

Property 5.1 (Reality of reputation). A reputation system is exactly as real as the identity it keys on. If an actor can mint a fresh identity at zero cost, then every sanction is escapable by respawn (**whitewashing**) and every quorum is forgeable by replication (**Sybil**), and the reputation reduces to a per-incarnation scratchpad with no economic content.

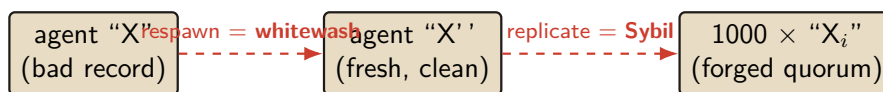
Today, Port Daddy identities are *self-asserted strings*: the `project:stack:context` triple (`lib/actor-roster.` is operator-chosen and re-pickable. That is fine for single-operator legibility (Papers 1–2) and

fatal for reputation. The fix is the one genuinely architectural piece of the L3 program: a daemon-minted, non-forgeable `actor_id` bound to a credential the actor cannot re-pick.

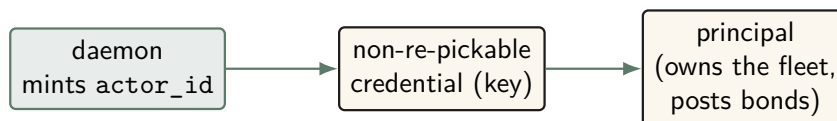
5.1 The classic attacks this forecloses

- **Cheap-pseudonym whitewashing** [6]: if a fresh identity is free, a bad record is shed by respawn. The defense is a *newcomer policy as a shape, not a scalar*: a newcomer may have full *work* ability but a reduced *economic ceiling* until it has a witnessed history — so identity churn costs more than a clean record is worth.
- **The Sybil attack** [7]: free identities defeat any reputation or quorum mechanism by replication. The defense is a daemon-minted id whose creation is rate-limited and, in the economy, *bonded*.

Self-asserted id (`project:stack:context`) — both attacks are free:



Daemon-minted non-forgeable `actor_id` (ADR-0040), bound to a principal:



Newcomer policy = a shape, not a scalar. A fresh id may have full *work* ability but a reduced *economic ceiling* until it earns a witnessed history (Friedman–Resnick). Identity churn then costs *more* than a clean record is worth, and a sanction *survives* a respawn.

Figure 6: **The non-forgeable root forecloses the two classic attacks.** With self-asserted strings (top), whitewashing and Sybil are free. A daemon-minted `actor_id` bound to a non-re-pickable credential and a principal (bottom) makes both costly: churn costs more than a clean record, and sanctions survive respawns. This is the one genuinely architectural piece of the L3 program.

5.2 The principal above the actor

The economic counterparty in every trade is not the agent but the **principal** — the human or organization that owns the fleet and stands behind its bonds. Bonds, reputation inheritance, and sanctions must ultimately key on the principal via the delegation chain, or Sybil bond-farming works by spawning fresh *agents* under one principal. This paper defines the principal as the delegation chain’s terminus and the identity object on which Paper 4’s market counterparty is built; Paper 4 *cross-references* this definition rather than redefining it.

Definition 5.1 (Principal). A **principal** is the root of a delegation chain: the durable economic entity (human/org) that owns a fleet, posts its bonds, and inherits the reputational consequence of its agents’ witnessed outcomes. Each agent’s `actor_id` is bound to exactly one principal at mint time.

Exercises

Check. (1) State Property 5.1 in your own words and give the two-attack proof sketch.

Trace. (2) Why is a “newcomer policy as a shape” (full work, reduced ceiling) strictly better than “newcomers are distrusted” (reduced work)? Consider the cost to an honest newcomer vs. a whitewasher.

★ *Open.* (3) Bond-farming: a single principal spawns 1000 agents, has them trade trivial tasks among themselves to mint reputation, then sells the “experienced” agents. Which of {principal binding, listing fee, sampled adversarial re-audit of high-velocity counterparty pairs} defeats this, and at what false-positive cost to honest high-volume pairs?

6 The keystone split: ADR-0040 vs. ADR-0040b

The non-forgeable identity of §5 is the volume’s “highest-leverage unbuilt keystone.” But the keystone is *two stones*, and conflating them is the single most consequential error the volume can make.

Definition 6.1 (ADR-0040, identity-local). **ADR-0040 (identity-local)** (docs/adr/0040-non-forgeable-a provides non-forgeable actor identity *within one trusted daemon*. Its threat model is explicitly intra-fleet, single-operator, and explicitly *not* a PKI; it offers no defense against a hostile *operator*. This is exactly what L0/L1/L2 — Papers 1, 2, and *this* paper — need and assume. **DESIGNED**.

Definition 6.2 (ADR-0040b, cross-operator attestation). **ADR-0040b (cross-operator attestation)** is the *unwritten* problem of binding `actor_id` keys across harbors you do *not* own: an actual key-distribution / attestation story between mutually-distrusting operators. A federated witness log gives log-*integrity*, not identity-*binding*. This is what Paper 4’s market needs and does *not* yet have. **VISION**(named gap, owned by Paper 4).

This paper depends on ADR-0040 (identity-local) and hands Paper 4 the unsolved ADR-0040b (cross-operator attestation) as a named dependency — not an assumption. Every L3 sentence that says “the boundary neither operator controls” is resting on ADR-0040b; until ADR-0040b exists, it is resting on a single trusted root, and we say so.

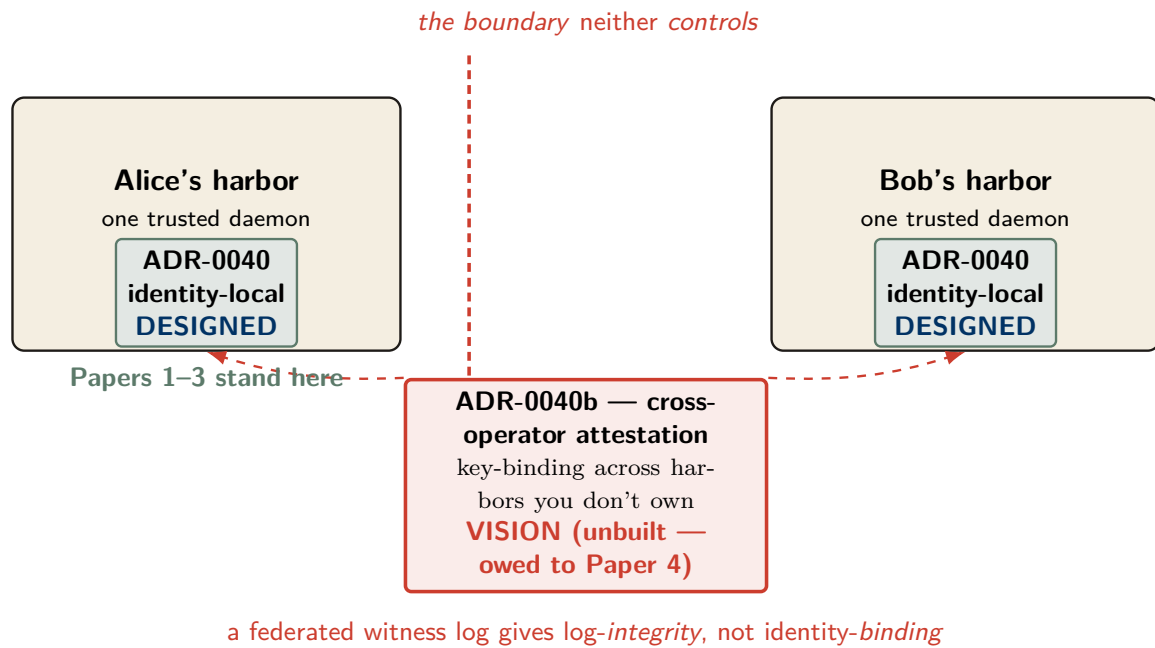


Figure 7: **The keystone is two stones.** ADR-0040 (identity-local) secures identity *within one trusted daemon* — exactly what Papers 1–3 need and assume. The cross-operator boundary needs ADR-0040b (cross-operator attestation), which is unwritten. This paper depends on the solid stone and hands Paper 4 the unbuilt one *by name*, rather than borrowing ADR-0040's single-operator threat model across the boundary.

Key idea. The keystone that holds up *this* paper (identity inside one trusted daemon) is *not* the keystone that holds up the cross-operator market. Paper 3 stands on ADR-0040. Paper 4 needs ADR-0040b and must declare it **VISION** rather than borrow ADR-0040's single-operator threat model. This is the clean boundary the volume's cross-layer review demanded.

Exercises

Check. (1) In one sentence each, state what ADR-0040 provides and what ADR-0040b would have to add.

Trace. (2) Take any claim in Paper 4 of the form "Alice's harbor and Bob's harbor settle across a boundary neither controls." Which stone does it secretly assume?

What is the actual security guarantee if only ADR-0040 exists?

★ *Open.* (3) Can a federated witness log ever substitute for identity-binding? Argue why log-integrity $\neq$ key-binding, then sketch the minimal additional assumption (a shared root of trust? a sponsor chain? a transparency log) that would close the gap, and the cost each imposes.

7 Reputation as a substrate, not a score

Here is the hinge of the whole volume, stated as the canonical cross-paper sentence:

The reputation estimator is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.

A team tempted to start an agent economy by “building an Elo leaderboard for backends” would be building the cheap, last part first. Elo, Bradley–Terry, and TrueSkill are well-understood; you can implement any of them in an afternoon over a table of game results. The hard, prior, load-bearing work is the three links *beneath* the estimator: a non-forgable identity (§5), continuity (§4), and an *oracle-witnessed* outcome ledger (Def. 4.1). Without those, the estimator scores noise — or worse, scores a number an adversary controls.

7.1 The estimator family (the well-understood part)

We catalog the estimators not because they are hard but because choosing the *wrong* one for the signal type is a real error.

- **Pairwise / tournament signal** → **Bradley–Terry / Elo / TrueSkill**. When the witnessed signal is “*A* beat *B* on this task,” the **Bradley–Terry** model [8] is the full-history MLE, and **Elo** [9] is its online special case. A SQLite harbor holds the *whole* history of every settled outcome, so it lives in the Bradley–Terry regime, not the streaming-Elo regime; BT gives more stable ratings than Elo’s recency-weighted update when the population is static. **TrueSkill** [10] adds a calibrated uncertainty σ — the principled way to say “we have not tested this backend much yet,” which matters at cold start.
- **Trust-propagation signal** → **EigenTrust**. When the signal is “who does the network trust,” **EigenTrust** [11] computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation-systems analogue of PageRank, and the canonical answer to “aggregate peer opinions while resisting collusive cliques.” It is the right tool for the *federated* reputation Paper 4 needs (trust that travels across harbors), and it is *not* a bandit.
- **Marketplace feedback signal** → **feedback systems**. Resnick–Zeckhauser [12] measured a $\sim 8\%$ reputation price premium on eBay; Tadelis [13] frames feedback systems as the platform’s answer to adverse selection. This is the empirical evidence that the third side of the market has real value to price.

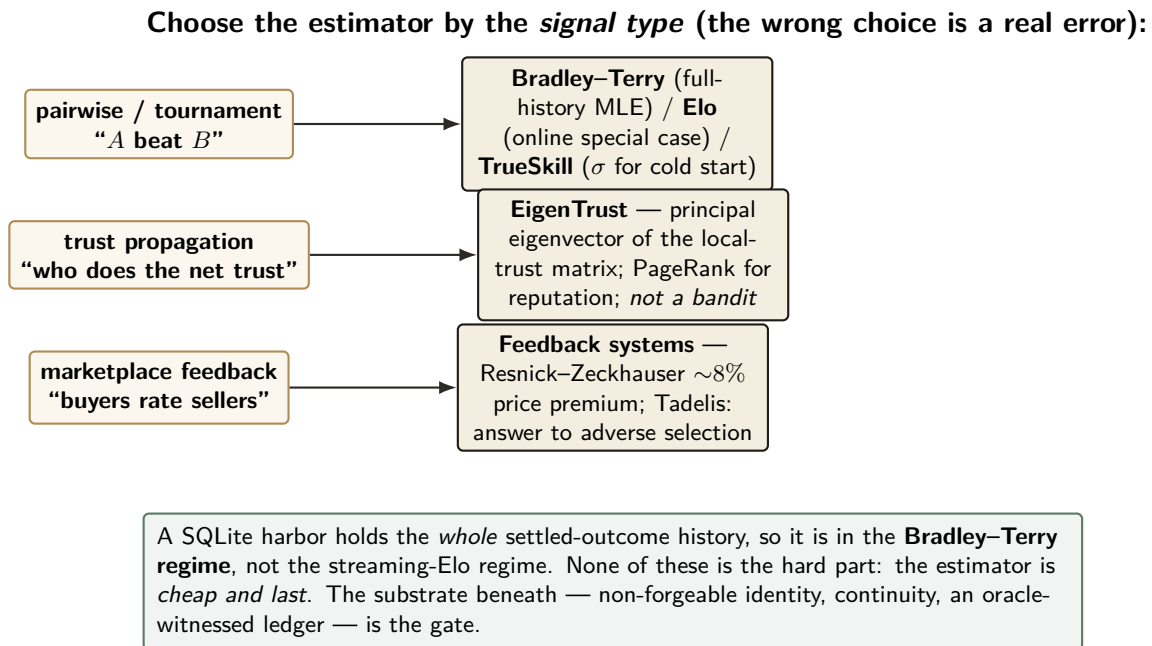


Figure 8: **The estimator family, by signal type.** Pairwise outcomes call for Bradley-Terry / Elo / TrueSkill; network trust calls for EigenTrust; marketplace feedback calls for feedback systems with measured price premia. All are well-understood and quick to build — which is exactly why they are the cheap, last part of the program.

7.2 Score as telemetry; gate on predicates

Key idea. Publish the score as *telemetry* (a signal the operator and the buyer can read), but *gate* on *predicates* (machine-checkable witnessed facts: “passed the acceptance test,” “no slashed bond in the last k settlements”). The instant the score itself becomes the gate, it becomes a Goodhart target [19]: “when a measure becomes a target it ceases to be a good measure.”

Exercises

Check. (1) Why is a SQLite harbor in the Bradley-Terry regime rather than the streaming-Elo regime?

Trace. (2) You have one task with a clear binary acceptance test, and another judged only by a human’s aesthetic preference. Which estimator fits each, and why does EigenTrust fit neither directly?

★ *Open.* (3) The substrate-vs-score sentence says the estimator is “cheap and last.” Construct a scenario where a *better estimator* still buys nothing because the substrate is rotten (e.g. forgeable identity or an agent-authored oracle). Generalize the lesson.

8 Why reputation is *not* a bandit problem

It is tempting — and the routing literature invites the temptation — to model “which agent/model/skill should I pick?” as a **multi-armed bandit** [15]: arms are agents, pulling an arm yields a reward, and Thompson sampling [16] or UCB balances exploration against exploitation. That framing is correct for one narrow internal task and catastrophically wrong for reputation as a whole.

The distinction is load-bearing enough to state as a claim.

Internal routing — *one operator privately choosing which of their own agents to spawn next* — is *bandit-shaped*. Reputation — *a public, adversarial, multi-dimensional, persistent substrate other parties trade on* — is *not*. Modeling reputation as a bandit imports four assumptions that are all false in a market.

Bandit assumes...	...but reputation is...	The right tool instead
Stationary arm rewards	non-stationary: models upgrade, skills version, reputation bubbles build & collapse (Liu-Skrzypacz)	bounded-memory reputation dynamics; version-binding
One scalar reward	multi-dimensional: accuracy / aesthetics / efficiency, in tension, buyer-weighted	a quality <i>vector</i> + buyer preference (ADR-0049)
Non-strategic environment	strategically adversarial: Goodhart, wash-trading, white-washing, collusion	mechanism design (Nisan et al.), bonds & slashing
Observed reward signal	produced by a capturable oracle with its own incentives	bond-and-slash judges; rate-the-raters (§10)

What survives the bandit framing: cold-start *exploration*. TrueSkill's σ and a UCB bonus are the same idea — “test the under-measured arm.” So *one operator's private routing* (choosing which of *their own* agents to spawn next) is legitimately bandit-shaped. The error is exporting that frame to the *public substrate*, where all four breaks bite at once.

Figure 9: **Why reputation is not a bandit problem.** A multi-armed bandit imports four assumptions — stationarity, a scalar reward, a non-strategic environment, and an observed reward signal — all false for a public reputation substrate. Exploration (one operator's private routing) is the one place the bandit framing legitimately survives.

8.1 The four broken assumptions

1. **Stationarity.** Bandit regret bounds assume the reward distribution of each arm is (near-)stationary. Agent quality is *non-stationary* by construction: models are upgraded, skills are versioned, prompts drift, and a skill's reputation built on v1 says nothing about v2. The Liu-Skrzypacz line of work on *reputation bubbles* [17] shows that with bounded memory and non-stationary types, reputation can build and *collapse* in cycles — behavior a stationary-arm bandit cannot represent.
2. **One scalar reward.** A bandit optimizes a single scalar. Quality is *multi-dimensional*: accuracy, aesthetics, and efficiency are different axes, often in tension (the fast cheap answer is rarely the most beautiful), and a buyer's weighting of them is a *preference vector*, not a constant. Collapse them into one scalar and you have re-created the Goodhart target the substrate argument warned against (§9).
3. **A non-strategic environment.** Bandit arms do not *try* to look good. Agents and their principals are *strategic adversaries*: they will Goodhart the measured axis, wash-trade fake outcomes, whitewash a bad record, and collude. A reputation mechanism is a *game*, and the relevant tool is mechanism design [18], not regret minimization.

4. **A trusted reward signal.** The deepest break: a bandit *observes* its reward. In a market, the reward (the grade) is *produced* by an evaluator who has their own incentives. The grading oracle’s incentive-compatibility is not a side-issue — it is a hole in the core theorem (§10). A bandit framing simply assumes the hole away.

8.2 What survives the bandit framing

To be fair to the bandit literature: cold-start *exploration* is real, and the right place to borrow from bandits is the *exploration bonus*. TrueSkill’s σ and a UCB-style bonus are the same idea — “test the under-measured arm.” So *within one operator’s private routing*, a contextual bandit conditioned on the operator’s cost/quality preference vector is a fine engineering choice. The error is exporting that frame to the *public substrate*, where non-stationarity, multi-dimensionality, strategy, and an untrusted reward signal all bite at once.

Pitfall. “We’ll just run Thompson sampling over the agents” is the most common way an agent-economy design quietly assumes its hardest problems away. It assumes a trusted reward (no judge-capture), a scalar reward (no aesthetics-vs-efficiency tension), a non-strategic environment (no Goodhart, no wash-trading), and stationarity (no model upgrades). All four are false in a market.

Exercises

Check. (1) Name the four bandit assumptions and the market reality that breaks each.

Trace. (2) Identify the *one* place a bandit *is* the right tool in this paper, and explain why it survives the four objections there.

★ *Open.* (3) Chiu-Zhang-van der Schaar [21] show competitive agents over-invest in self-improvement, burning surplus despite individual rationality. Does capping the reputation signal’s marginal return restore efficiency, or merely relocate the waste? This is the self-improvement arms-race open problem.

9 ADR-0049: multi-dimensional reputation and neutral judges

We now design, in this paper, the architecture the volume calls **ADR-0049**. It has three parts: a multi-dimensional quality vector; a market of neutral, conflict-free, influence-free evaluators; and a skill→agent helpfulness attribution. We state it as a design (**DESIGNED-to-VISION**), honestly: no `lib/reputation.ts` exists yet.

9.1 Multi-dimensional quality (different axes, different judges)

Definition 9.1 (Quality vector). A witnessed outcome carries a **quality vector** $\mathbf{q} = (q_{\text{acc}}, q_{\text{aes}}, q_{\text{eff}})$ over the axes *accuracy* (did it do the right thing, against an oracle), *aesthetics* (is the artifact good — code clarity, design taste — judged by an expert), and *efficiency* (tokens, wall-clock, dollars, from the cost ledger `lib/cost-ledger.ts`, **BUILT**). Each axis is scored by a *different judge*, because the competent judge of accuracy (a test oracle) is not the competent judge of aesthetics (a senior reviewer) is not the competent judge of efficiency (a meter).

The buyer reads the *vector* and applies their own weighting (the operator’s “cheap vs. careful” preference). Publishing a vector and letting the buyer weight it is the natural disclosure form — and it is precisely what avoids re-collapsing quality into one Goodhart-able scalar. The open

design question of *whether* a vector disclosure lets agents Goodhart the most-weighted axis is named as a starred exercise.

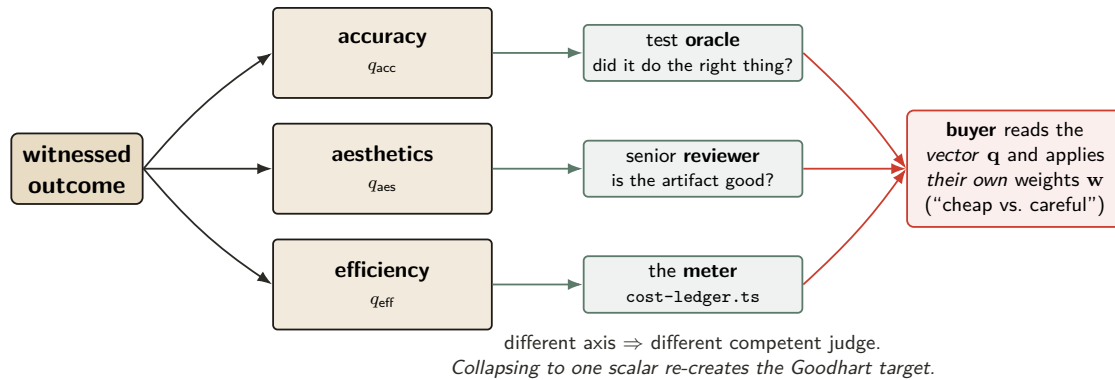


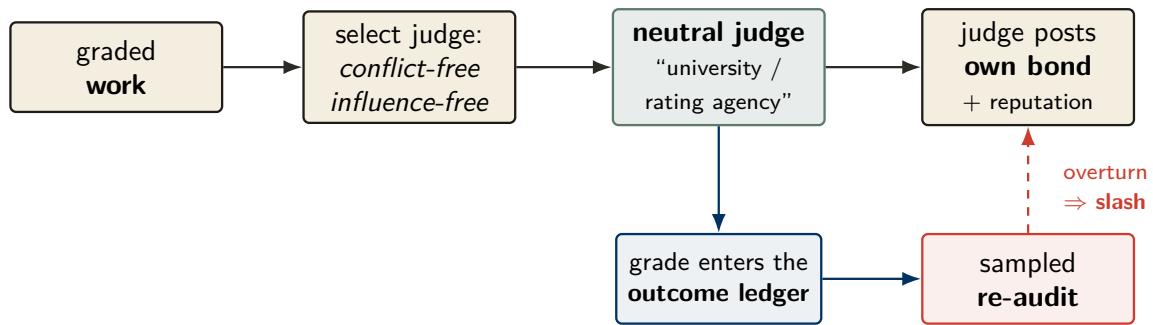
Figure 10: **Multi-dimensional reputation (ADR-0049)**. A witnessed outcome carries a quality *vector* $\mathbf{q} = (q_{acc}, q_{aes}, q_{eff})$. Each axis is scored by a *different* competent judge: a test oracle for accuracy, a senior reviewer for aesthetics, the cost meter for efficiency. The buyer reads the vector and applies their own weights — which is exactly what avoids collapsing quality into one Goodhart-able scalar.

9.2 The neutral-judge market (the harbor’s universities and rating agencies)

A judge that grades the work of a party it has a stake in is not a judge. The metaphor the volume uses is exact: a reputation needs the equivalent of *universities* (which certify competence) and *rating agencies* (which rate counterparties) — third parties whose value is precisely that they are *neutral*. We make “neutral” a checklist, not a vibe:

- **Conflict-free.** The judge has no bond, stake, or principal-link in the work being graded. (Excludes self-grading and same-principal grading.)
- **Influence-free.** The judge cannot be paid *by the graded party* for a better grade. Funding is the hard part: a paid judge has a client, an unpaid judge does not scale. The resolution (§10) is that judges post their *own* bond and carry their *own* reputation.
- **De-biased, for LLM judges.** An LLM-as-judge [14] carries position, verbosity, and *self-enhancement* bias (a backend rates its own family up). The discipline is blind, order-swap, pairwise, and family-exclude scoring — never let a backend judge its own family unblinded.

Crucially, an expert third-party judge is *hired through Port Daddy’s own protocols*: judging is itself a float-planned, bonded task (Paper 4’s escrow ceremony), so the harbor that runs the market also runs the hiring of the judges who keep it honest. Figure 11 draws this: the graded work flows to a judge selected for conflict-/influence-freedom, the judge posts a bond, the grade enters the outcome ledger, and a later re-audit can slash a judge whose grade is overturned.



The judge is **hired through Port Daddy's own protocols**: judging is itself a float-planned, bonded task (Paper 4's escrow ceremony). The harbor that runs the market also runs the hiring of the judges that keep it honest — and a judge whose grade is later overturned is *slashed*, so the judge's own reputation is the asset it protects.

Figure 11: **The neutral-judge market.** Graded work flows to a judge selected for conflict- and influence-freedom; the judge posts its own bond, the grade enters the outcome ledger, and a sampled re-audit can slash a judge whose grade is overturned. “Neutral” is a checklist (conflict-free, influence-free, de-biased), not a vibe, and the judges are hired through the harbor’s own bonded protocols.

9.3 Skill → agent helpfulness attribution

A distinctive, tractable, and *shippable-soon* signal: track whether a **skill** (a `SKILL.md`, its referenced docs, or its scripts — `skills/*/SKILL.md`) actually *helped* an agent, *especially when that skill's content was loaded into the agent's context window*. The witnessed event is concrete: skill s was grafted into agent a 's context at task t ; t later settled with quality vector $\mathbf{q}$. Over many such events we estimate a *skill-helpfulness* effect — did loading s raise $\mathbf{q}$ relative to comparable tasks without it? This is reputation for *tools*, keyed on witnessed, context-attributable outcomes, and it is the licensing side's substrate (Paper 4). It is also the cleanest near-term instance of the whole thesis: a non-forgable event (the graft), a witnessed outcome (the settled task), and a multi-dimensional grade — no bandit required.

Pitfall. Attribution is not causation. “Skill s was in context and the task succeeded” does not prove s caused the success. The honest estimate needs a comparison set (similar tasks without s) and is still observational. Sell it as *helpfulness evidence*, not *proof of value*, and gate licensing clawbacks on the witnessed green/red settlement, not on the helpfulness estimate alone.

Exercises

Check. (1) For each of accuracy / aesthetics / efficiency, name a competent judge and an *incompetent* one.

Trace. (2) Walk a judging task through Port Daddy's bonded protocol: who float-plans it, who posts the bond, where does the grade land, and what slashes the judge?

Trace. (3) Design the witnessed event for skill-helpfulness. What fields does it need so that a buyer can verify “this skill helped” without trusting the skill's author?

★ *Open.* (5) If you publish the 3-vector and buyers weight it, do agents Goodhart the most-weighted axis? If you publish a scalar, you have recreated the bandit framing the substrate argument rejects. What is the right disclosure form for vector reputation?

10 The grading oracle's incentive-compatibility

Every folk-theorem incentive-compatibility result in the L3 economy bottoms out in “the daemon derives the grade” — but the grade is produced by a *capturable evaluator*. This is not a side-gap. It is a hole in the *core theorem*, and we treat it head-on rather than assuming it away.

Property 10.1 (The grading-oracle hole). Let an outcome's settlement (and thus the bond flow and the reputation update) depend on a grade g produced by an evaluator J . If J can be captured — bribed, colluded with, or self-interested — then the incentive-compatibility of *every* mechanism that settles on g is only as strong as J 's own honesty. Assuming J honest is assuming the conclusion.

There are two honest ways to close the hole, and a recursion to bound.

Option A — bond-and-slash the judges. Make judging a bonded role (§9): J posts a bond before grading; a later **re-audit** (a fresh, conflict-free judge, sampled) that overturns J 's grade *slashes* J . Now lying has an expected cost, and a judge's own reputation is the asset they protect. This converts “trust the oracle” into “the oracle has skin in the game,” which is the same move slashing makes for agents.

Option B — 2-of-3 multi-oracle. Settle disputes against a quorum of independent oracles (automated test / evidence review / human), so capturing the grade requires capturing a majority. This raises the cost of capture but does not eliminate it, and the *human* oracle is a scarce, expensive, capturable resource in its own right (the economics of arbitration capacity is a named open problem, below).

The rate-the-raters recursion — does it terminate? Option A pushes the problem up a level: who audits the auditors? The re-auditor is itself a judge with incentives. The recursion “rate the raters who rate the raters” must be shown to *terminate* or to *contract*. We do not claim a proof; we claim the shape of one and mark the rest open.

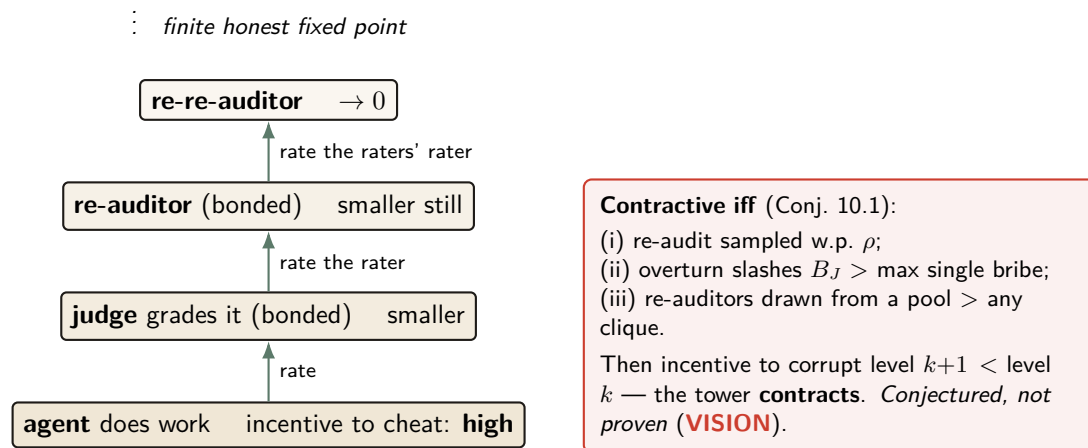


Figure 12: **The rate-the-raters recursion.** Bond-and-slash pushes the grading problem up a level: who audits the auditors? Under sampling, a slash exceeding the maximum bribe, and a conflict-free re-auditor pool larger than any clique, the incentive to corrupt each level is strictly smaller than the level below — making the tower contractive with a finite honest fixed point. This is the precise statement of what Paper 4's central IC theorem needs and currently lacks.

Conjecture 10.1 (Contractive re-audit). If (i) re-audits are sampled with probability ρ , (ii) an overturned grade slashes a bond B_J strictly exceeding the maximum bribe a single graded

party can offer, and (iii) re-auditors are themselves drawn conflict-free from a pool larger than any single colluding clique, then the expected gain from dishonest grading is negative at *every* level of the recursion, and the “rate-the-raters” tower is *contractive*: the incentive to corrupt level $k + 1$ is strictly smaller than at level k , so the tower has a finite honest fixed point.

Pitfall. Conjecture 10.1 is *conjectured*, not proven. It is the precise statement of what Paper 4’s central incentive-compatibility result *needs* and currently *lacks*. The contribution here is to name the hole exactly, give two candidate closures, and state the termination obligation — *not* to claim it is closed. Honest labels apply: this is **VISION**.

Key idea. The grading oracle is the load-bearing assumption hidden under every IC claim in the economy. “The daemon derives the grade” only helps if the grade’s *producer* is itself incentive-compatible. Either bond-and-slash the judges and prove the rate-the-raters recursion contracts, or accept that the core theorem is conditional on an honest oracle — and say which.

Exercises

Check. (1) State Property 10.1 and explain why “the daemon derives the grade” does not, by itself, close it.

Trace. (2) Walk Option A: a judge grades dishonestly, a re-audit overturns it. Follow the bond flow and the reputation update for the judge.

★ *Open.* (4) Prove or refute Conjecture 10.1. If the re-audit probability ρ is too low, the tower is not contractive — find the critical ρ^* as a function of bond size and bribe ceiling.

★ *Open.* (10) Arbitration capacity: human oracles are scarce. Design a market for bonded, rated, slashable arbiters and argue whether it converges to honest rulings or to rubber-stamping under load.

11 Revocation, tombstones, and bounded memory

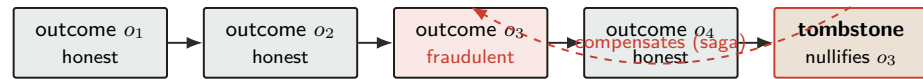
A reputation built on witnessed outcomes must answer two questions the cheerful “reputation never ends” framing dodges: what happens when a witnessed outcome turns out to be *fraudulent*, and is no-decay actually *good*? We adopt the volume’s reconciled position on both, against the naive version.

11.1 Monotone in honest outcomes, individually revocable

Property 11.1 (Reputation monotonicity with tombstones). Reputation is **monotone in honest witnessed outcomes** — a genuine delivery is never silently erased by the passage of time — *but* a witnessed outcome is itself **individually revocable via a propagating tombstone**: a compensating, append-only entry that retracts a specific outcome later found fraudulent (a colluding judge, a faked oracle). “Never decays” applies to honest outcomes *only*; it is *not* an anti-revocation property.

This is a **saga**-style compensation [20]: you do not mutate the ledger in place (that would break append-only and auditability); you append a tombstone that nullifies the targeted entry, and the tombstone *propagates* across harbors bounded by the federation’s revocation-convergence window (Paper 4). The poisoned data point was spendable only during the propagation window; bounding that window is the federation’s job.

Append-only outcome ledger (never mutate in place):



Property. Reputation is *monotone in honest outcomes* (o_1, o_2, o_4 never silently erased) but *individually revocable*: the fraudulent o_3 is nullified by an appended **tombstone**, which *propagates* across harbors bounded by the revocation-convergence window (Paper 4). The poisoned point was spendable only during that window.

Bounded memory (Liu–Skrzypacz) is a *tunable parameter*, not a virtue: ∞ -memory is rarely optimal — it freezes types and removes the incentive to keep performing.

Figure 13: **Tombstone revocation on an append-only ledger.** A fraudulent outcome is not deleted (that would break append-only and auditability); it is nullified by an appended *tombstone* — a saga-style compensating entry — which propagates across harbors. “Reputation never ends” is corrected to: monotone in *honest* outcomes, individually revocable via tombstone, with bounded memory as a design parameter.

11.2 Bounded memory is a design parameter, not a virtue to assert away

The naive “no decay window an agent can outrun” framing sounds principled but is mechanism-design-hostile. Liu–Skrzypacz [17] show that with *bounded* memory, reputation can be *welfare-improving* relative to infinite memory — and that infinite memory (∞) is rarely optimal, because it freezes types and removes the incentive to keep performing. Bounded memory is a **tunable parameter**, and the right setting is an empirical/mechanism-design question, not a slogan.

Pitfall. “Reputation never ends” as a *virtue* is wrong twice over: it makes a fraudulent outcome permanent (needs the tombstone of Property 11.1), and it asserts ∞ -memory is optimal when the reputation-bubble literature shows it usually is not. Engage bounded memory as a *parameter*; do not assert no-decay as a feature.

Key idea. Two corrections to the cheerful framing: (1) reputation is monotone in *honest* outcomes but *individually revocable* via a propagating tombstone — a saga compensation, not in-place mutation; (2) bounded memory is a *design parameter* (Liu–Skrzypacz reputation bubbles), not an anti-feature to asser away.

Exercises

Check. (1) Why must revocation be a tombstone (append) rather than a delete (mutate) on the outcome ledger?

Trace. (2) A judge is later found to have colluded on five settled outcomes. Walk the tombstone for one of them: what is appended, what propagates, and what bounds the window the poisoned reputation was spendable?

★ *Open.* (8) Reputation-claim revocation with an append-only, bounded-memory ledger across N harbors: how do you prove the correction converged and bound the spendable window, without violating append-only? This is the cross-harbor analogue of capability revocation (Paper 4).

12 What this paper hands the market (Paper 4)

This paper is the bridge; its output is a precise hand-off to *The Harbor Economy* (Paper 4). We summarize what is delivered, what is borrowed, and what is explicitly owed.

Artifact	Label	Role in Paper 4
Role / person distinction (Defs. 2.1–2.2)	—	The third side of the market is a <i>tradeable person</i> , defined here.
Principal-as-economic-entity (Def. 5.1)	DESIGNED	Bonds and sanctions key on the principal; Paper 4 cross-references, not redefines.
The three continuity organs	BUILT/BUILT-WEAK/DESIGNED	Reputation keys on organ 3 (the outcome organ).
Outcome ledger / oracle (Def. 4.1)	DESIGNED	The witnessed-outcome substrate every settlement updates.
ADR-0049 reputation (Defs. 9.1)	DESIGNED → VISION	The multi-dimensional score the market reads; the licensing side's signal.
Neutral-judge market	DESIGNED → VISION	The “universities and rating agencies” that keep grades honest.
Grading-oracle hole (Prop. 10.1, Conj. 10.1)	VISION	<i>Owed back to Paper 4</i> : its central IC theorem is conditional on closing this.
ADR-0040b cross-operator attestation (Def. 6.2)	VISION	<i>Named dependency Paper 4 must build</i> ; not assumed here.
Tombstone revocation (Prop. 11.1)	DESIGNED	Reputation is revocable; convergence is Paper 4's federation job.

Table 2: The bridge's hand-off to Paper 4. Two items flow *back* to Paper 4 as named, unsolved obligations: the grading-oracle's incentive-compatibility (Conj. 10.1) and cross-operator attestation (ADR-0040b). The volume is one book precisely because these debts are named in the same words on both sides of the seam.

Build the harbor first. The trade comes when the ships do — but the thing that makes the trade possible (continuity that turns spawns into persons) is the same thing that makes the single-player tool good. The bridge is not a detour; it is the load path.

13 Open problems (the starred exercises, collected)

The genuinely unsolved problems this paper surfaces, gathered for the reader who wants the research frontier rather than the pedagogy:

1. **Branching continuity (§3)**. If a checkpoint is forked into two live incarnations, which inherits the reputation? Parfit says identity-is-not-what-matters; the economy needs a rule, and the rule invites an attack.
2. **Agent-death taxonomy (§4)**. What is recoverable vs. fundamentally lost when an LLM agent dies mid-thought? The hard half of checkpoint-with-teeth.
3. **ADR-0040b (§6)**. Cross-operator attestation: binding `actor_id` keys across harbors you do not own. The highest-leverage unbuilt keystone for the market.

4. **The grading-oracle / rate-the-raters recursion (§10, Conj. 10.1).** Prove the re-audit tower contracts, or accept the core IC theorem is conditional.
5. **Vector-reputation disclosure (§9).** Publish a vector and agents Goodhart the heaviest axis; publish a scalar and you have a bandit. The right disclosure form is open.
6. **Self-improvement arms race (§8).** Chiu et al. 2025: does capping reputation’s marginal return restore efficiency or relocate the waste?
7. **Reputation-claim revocation convergence (§11).** Surgically retract a fraudulent outcome across N harbors, prove convergence, bound the spendable window — without violating append-only.
8. **Bond-farming defense (§5).** Principal binding vs. listing fee vs. sampled re-audit of high-velocity pairs; the false-positive cost to honest high-volume counterparties.
9. **Arbitration capacity (§10).** A market for bonded, rated, slashable human/automated arbiters: honest rulings, or rubber-stamping under load?
10. **Skill-version reputation (§9).** A portfolio proof for skill v1 says nothing about v2; the lemons problem reappears at every version bump.

14 Conclusion

We set out to bridge the wedge to the market by answering one question: what has to be true before a coding agent’s work can be *sold*? The answer is not a score. It is a *self* — a role plus continuity, keyed on an identity that cannot be re-picked, accumulating a transitive, witnessed history that survives every respawn. The estimator that turns that history into a number is the cheap, last part; the substrate beneath it is the gate, and most of that substrate is honestly not built yet.

We were disciplined about the keystone (this paper stands on ADR-0040 and hands Paper 4 the unbuilt ADR-0040b by name), about the hardest hole (the grading oracle’s incentive-compatibility is a flaw in the core theorem, not a footnote), and about the cheerful slogans (reputation is monotone in *honest* outcomes but individually revocable, and bounded memory is a parameter, not a virtue). Reputation is not a bandit problem — it is non-stationary, multi-dimensional, strategically adversarial, and graded by an oracle with its own incentives.

A spawn does work and disappears. A person does work and stays — and only what stays can be priced. The harbor’s whole economy is the machinery for manufacturing, and then honestly grading, a self.

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